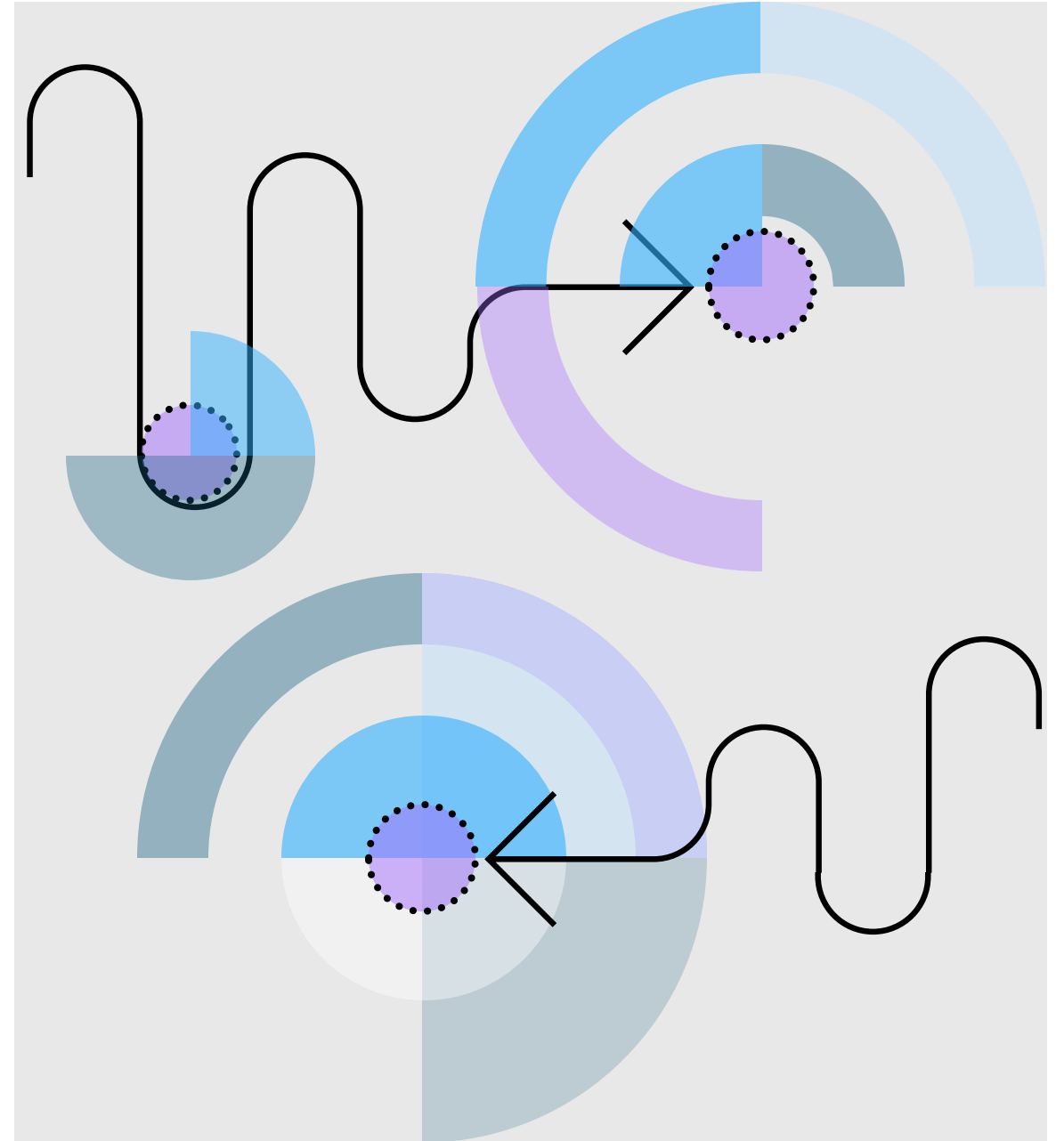


# AI Specialist Capstone project



# OBJECTIVE

This capstone project is your opportunity to move beyond theory and **apply AI in a real development workflow**. You will leverage **Agentic IDEs (such as IBM BOB)** to implement backend solution for an eCommerce application - just like in modern AI-powered engineering teams.

By the end, you won't just understand AI-assisted development - you'll be able to demonstrate it. Remember to focus not just on the outcome, but on how you leverage AI to think, build, and solve problems differently.

Follow these steps:

- Get an access to IBM BOB ([fill in this form](#)) (you can also use another Agentic IDE such as AWS Kiro)
- Install the IDE on your local setup ([Bob installation guidelines](#))
- Go through the use case from Slide 3 to Slide 11
- Complete the activities listed on Slide 12
- Join the slack channel [#applied-ai-specialist-capstone-project](#) for any peer or SME help

Note: Any software installed for development can be removed post the development

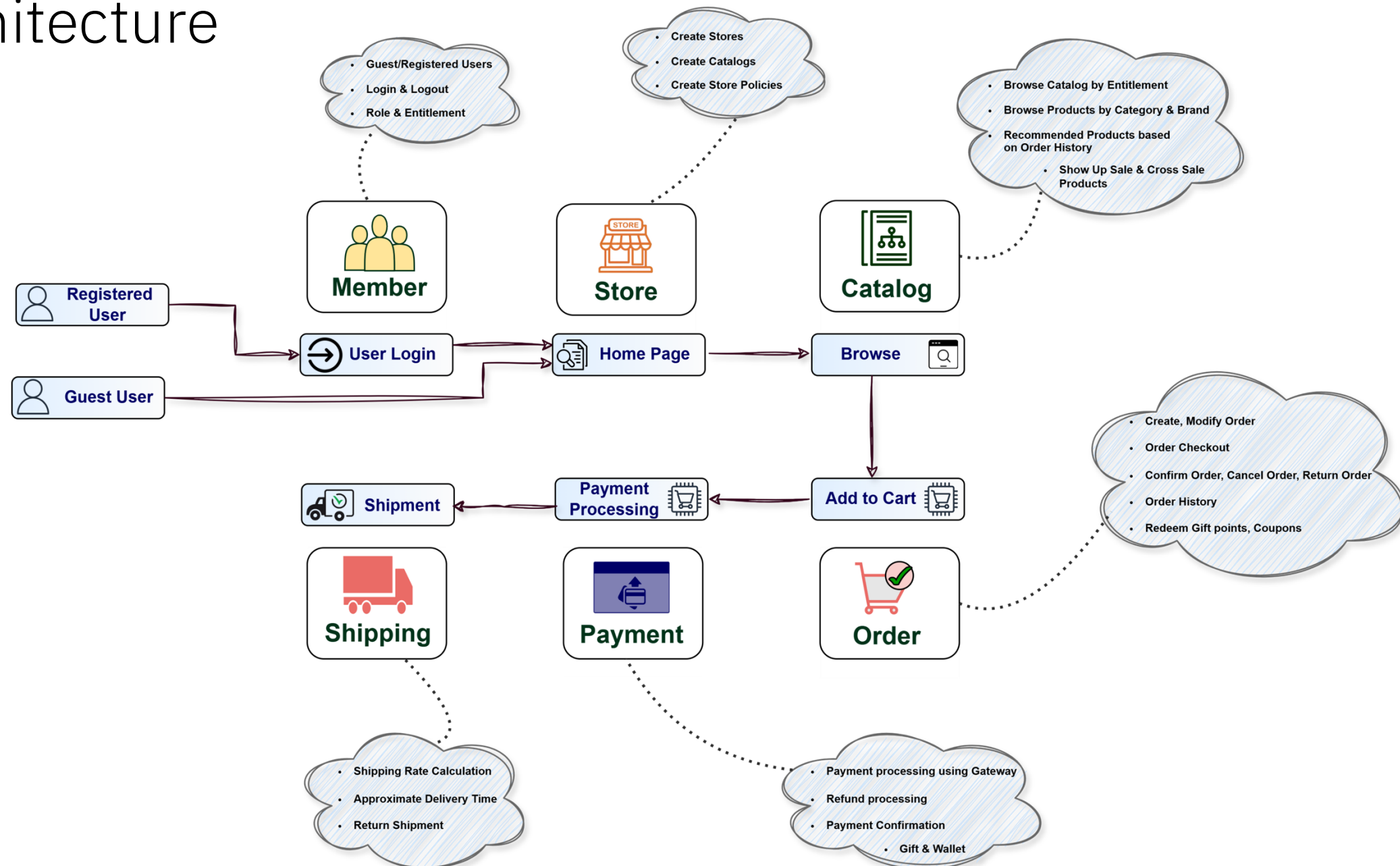
# USE CASE : An E-Commerce platform to view , browse , purchase books

**Description of use case :** An online store where a customer can browse , select and buy the selected products

The following are the customer journeys

| E-store Home |                                                      | Catalogue |                                                                  | Payment & Purchase confirmation |                                                        |
|--------------|------------------------------------------------------|-----------|------------------------------------------------------------------|---------------------------------|--------------------------------------------------------|
| 1            | Login Page                                           | 5         | Select the Category of products                                  | 9                               | Select the address for delivery                        |
|              | User Authntentication                                |           |                                                                  |                                 |                                                        |
| 2            | User Browses Order History with Buy In Again feature | 6         | Browse the brands                                                | 10                              | Initiate payment with the right option                 |
|              | Recommends items based on Order History              |           |                                                                  |                                 | Redeem gift points                                     |
| 3            | Access Product catalogue for each category           | 7         | Select the product (product tagged with tentative delivery date) | 11                              | Payment confirmation                                   |
|              |                                                      |           | Related products appear for selection                            |                                 |                                                        |
|              |                                                      | 8         | Add the product(s) to basket                                     | 12                              | Confirmation on purchase<br>Cancel Order Within 48 hrs |

# Architecture

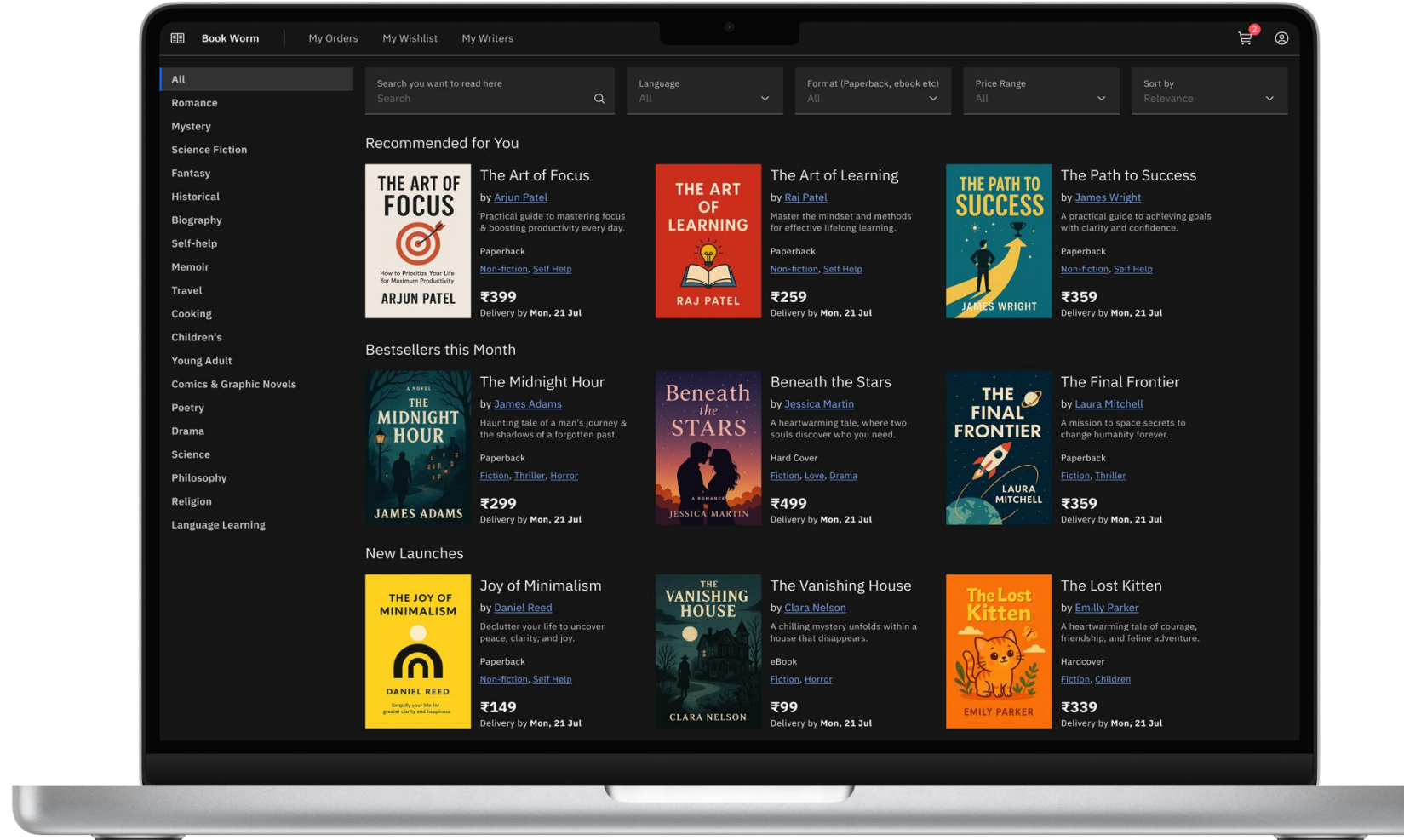


# E-Bookstore Platform to be build with agentic IDE

The following demonstrates an e-bookstore platform with the following characteristics:-

- Landing page with the availability of books
- Selection of books
- Buy the books and commit a payment

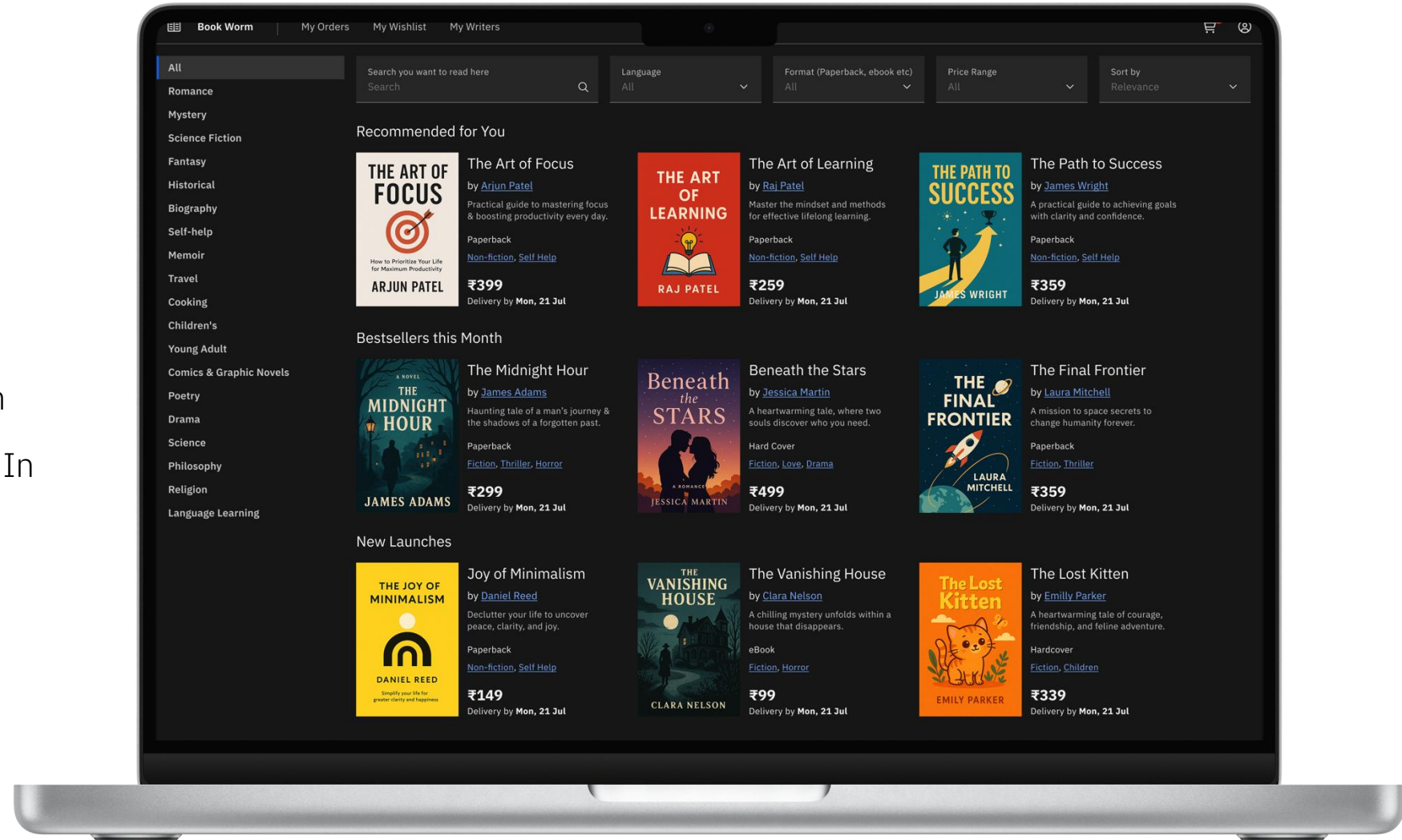
**\*\* The further slides will depict the build of the platform \*\***



# The Catalogue page of the e-bookstore is depicted as follows

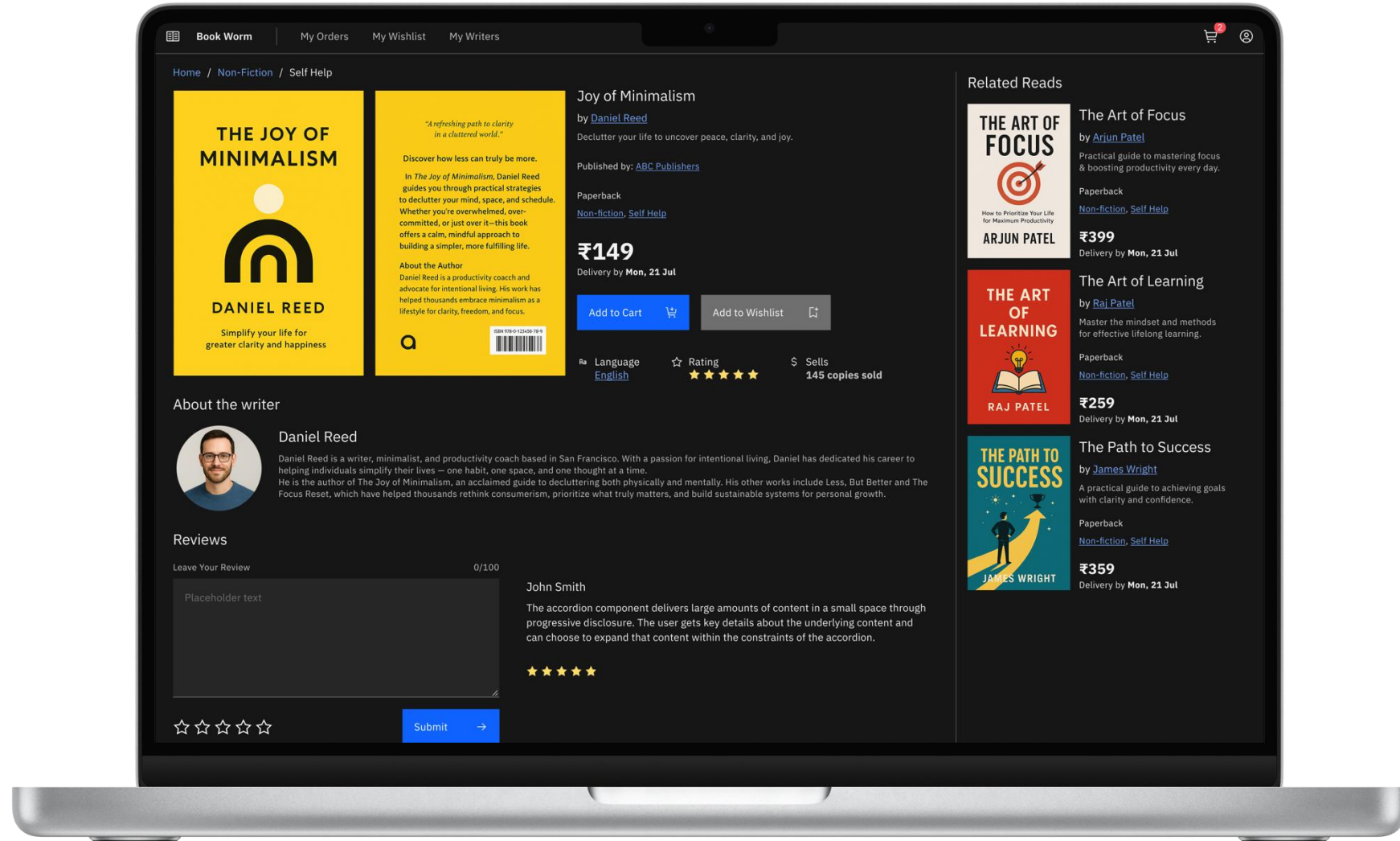
Key features:

- Select the Category of products
- Access Product catalogue for each category
- Browse the brands
- Select the product
- Related products appear for selection
- User Browses Order History with Buy In Again feature



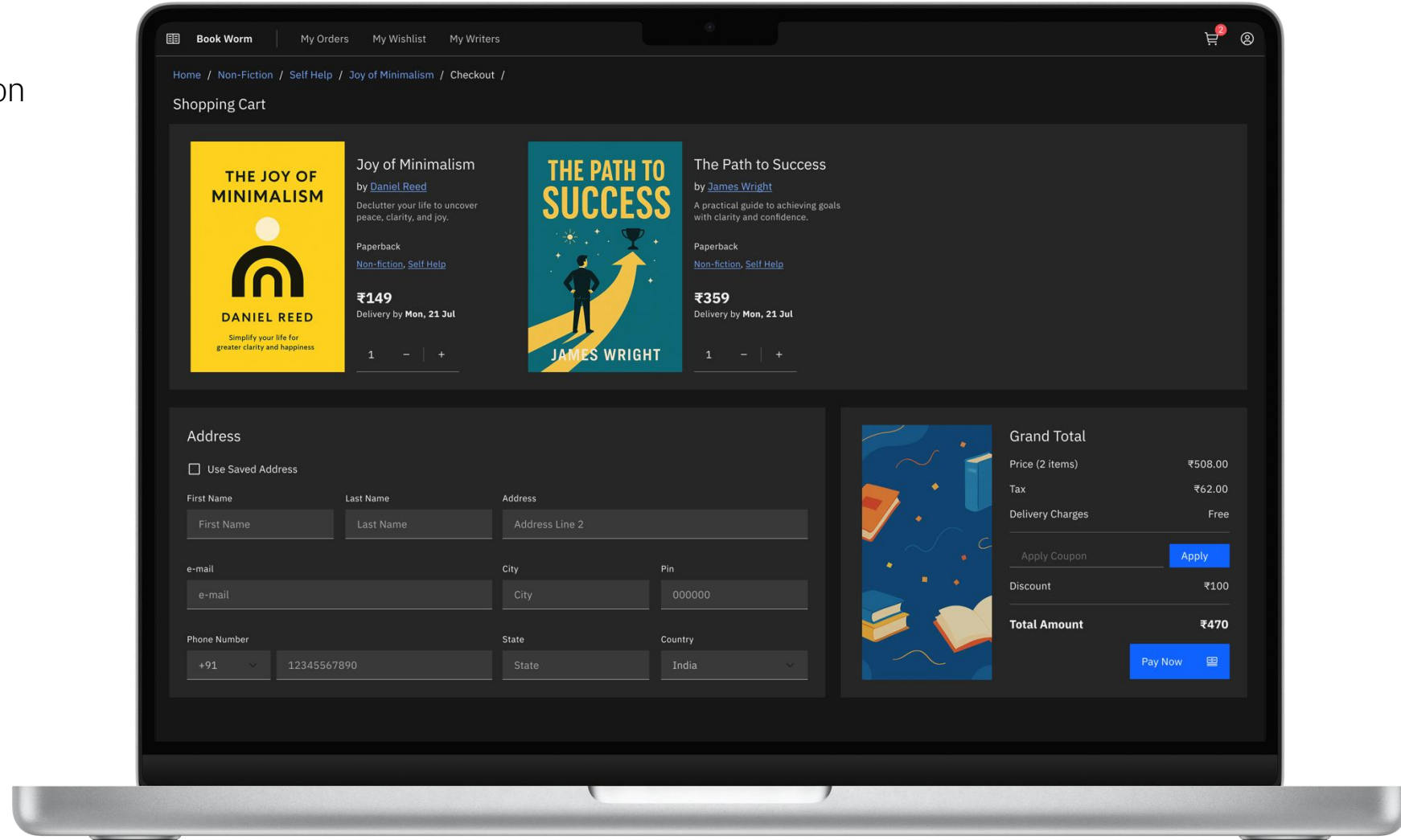
# The following provides the Shopping Cart for update

- Add the product(s) to basket
- Recommends items based on Order History



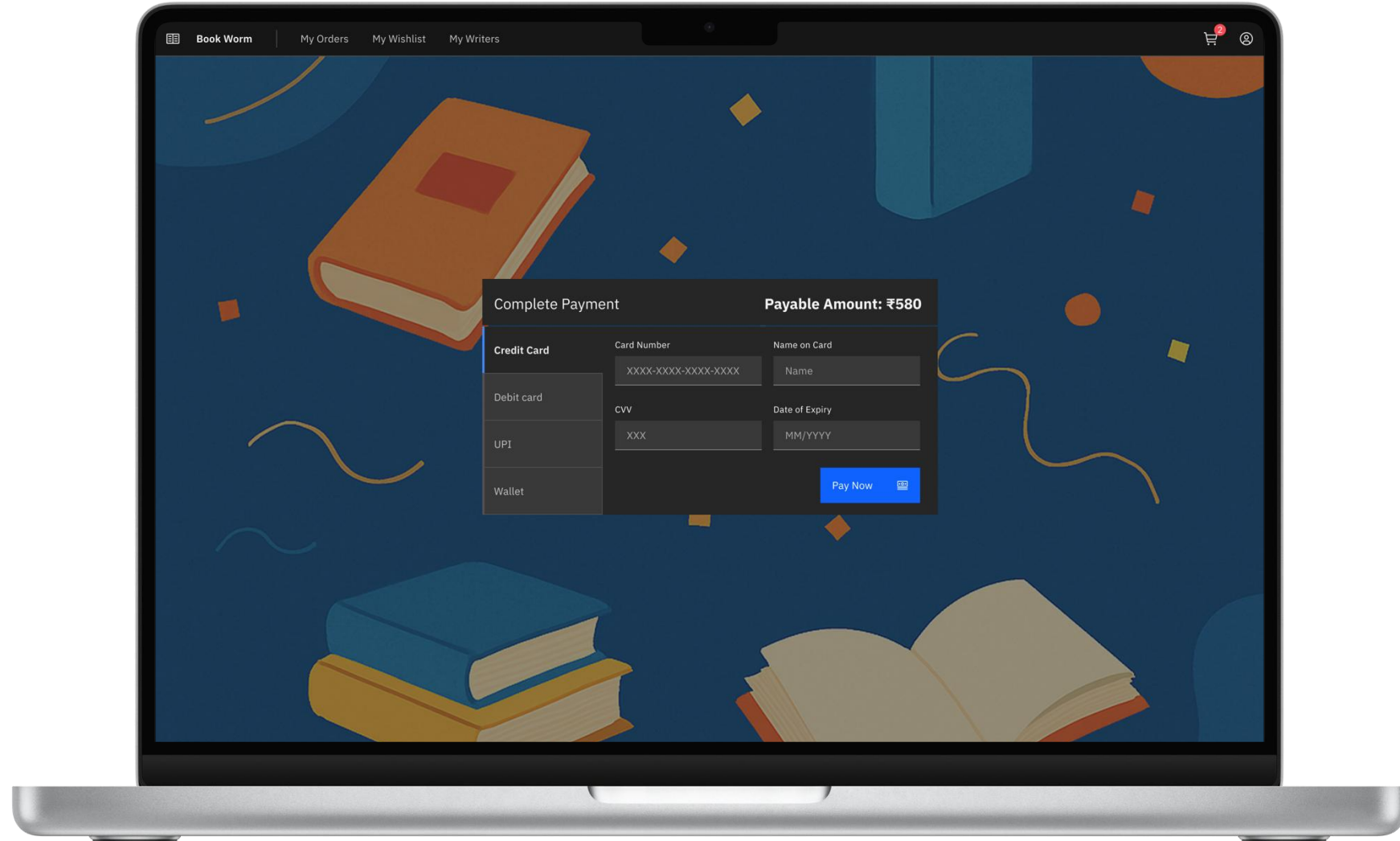
# Payment & Purchase

- Select the address for delivery
- Initiate payment with the right option
- Redeem gift points



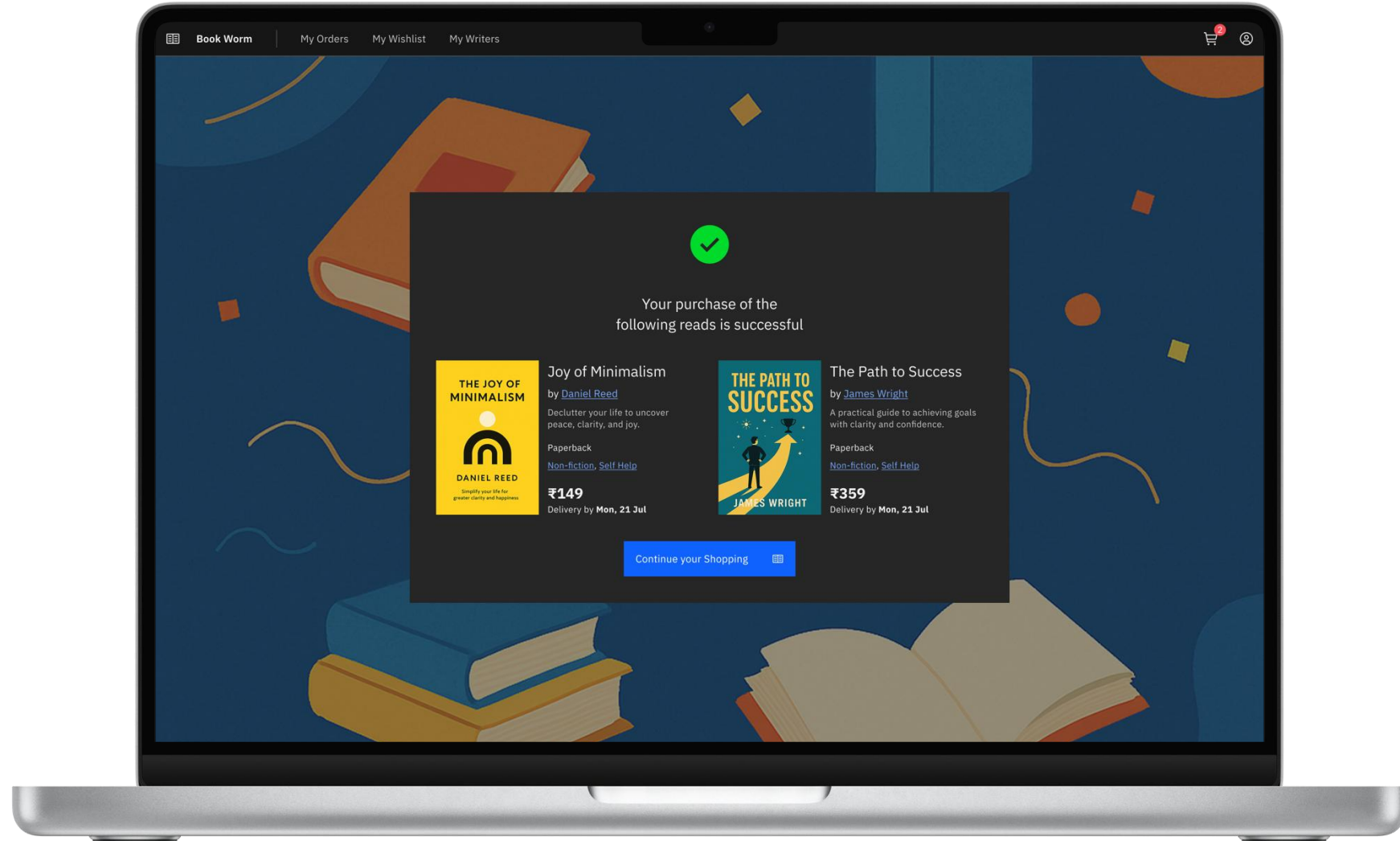
# Payment Screen

- Initiate payment with the right option (e.g., Credit card, Debit card)
- Complete the payment



# Payment & Purchase

- Confirmation on purchase
- The message should bring up the purchase completion



# Technology Components and Workflow

- Database: PostgreSQL
- AI augmentation on UI: Vibe coding with IBM BOB (or AWS KIRO)
- Prepare the data model design based on wireframes (Manual)
- Open API spec generation
- AI augmented backend services: IBM BOB (or AWS KIRO)
- AI augmented test case generator: ICA Agent (Choose an agent Ex: Test Scenario Generator, API Testcase generator, Test case generator for React etc..)
- Readiness and deployment on a local machine and if you have a cloud environment access (e.g AWS ROSA or IBM ROKS, you can deploy)
- PR on your personal GitHub with access provided to your Manager
- Record a video describing the steps and submit it to be reviewed by your Manager

# Backend API Development Workflow with IBM BOB/KIRO

## Step 1: Analyze Wireframe

- Open wireframe documents referred from slide 5- Slide 10
- Identify data entities (Product, User, Order, etc.)
- List required features (search, filter, add to cart, etc.)

## Step 2: Open IBM BOB/KIRO

- Launch IBM BOB or KIRO tool
- Start a new session for backend development

## Step 3: Generate API Specification

- Upload wireframe to BOB/KIRO
- Ask: "Generate OpenAPI specification for this e-commerce wireframe"
- Specify entities and operations needed
- Review and save the generated API spec

## Step 4: Generate Spring Boot Code

- Provide API spec to BOB/KIRO
- Ask: "Generate Spring Boot application with these APIs"
- Specify: Java version, database type (MySQL/H2), dependencies needed
- Download/copy generated code

## Step 5: Setup Project Locally

- Extract/organize generated code
- Open in IDE (IntelliJ, VS Code)
- Review project structure (controllers, services, repositories, entities)

## Step 6: Configure Database

- Install postgresql community edition from [PostgreSQL: Downloads](#) (as per the local OS)
- Update application.properties file
- Set database connection details
- Configure port and application name

## Step 7: Run Application

- Build project: ./mvn clean install
- Start application: ./mvn spring-boot:run
- Verify application starts without errors

## Step 8: Test APIs

- Open an API testing tool like Insomnia or through browser
- Test each endpoint (GET, POST, PUT, DELETE)
- Verify responses match API specification
- Check data is saved in database

## Step 9: Git Check-in

- Initialize git: git init (if new project)
- Create feature branch: git checkout -b feature/api-implementation
- Stage files: git add .
- Commit: git commit -m "Implement Spring Boot API for e-commerce"
- Push: git push origin feature/api-implementation
- Create Pull Request with API documentation

Raise any questions through the slack channel:

[#applied-ai-specialist-capstone-project](#)